

One Dimensional Heat Flow

Consider a rod of homogeneous material with a density of ρ (g/cm^3) and a constant cross-sectional area A (cm^2). It is assumed that the sides of the bar are insulated, meaning heat loss through conduction or radiation from the sides is negligible.

By taking one end of the bar as the origin and the direction of heat flow as the positive x -axis, let ' s ' be the specific heat and ' k ' be the thermal conductivity of the material.

If $u(x, t)$ represents the temperature at a distance x at time t , then the one-dimensional heat equation is given by:-

$$\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2} \quad \left[\text{where, } c^2 = \frac{k}{\rho s} \right]$$

$$\text{If } \frac{\partial u}{\partial t} = 0, \quad \frac{\partial^2 u}{\partial x^2} = 0$$

This eqⁿ. is called Laplace eqⁿ. in two dimensions.

Solution:

The one dimensional heat equation is,

$$\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2} \quad \text{--- (1)}$$

Let, $u = XT$ --- (2) be a solution of eqⁿ. (1).

$$\frac{\partial u}{\partial t} = X \frac{dT}{dt} \quad \& \quad \frac{\partial^2 u}{\partial x^2} = T \frac{d^2 X}{dx^2}$$

Putting in eqⁿ. (1),

$$X \frac{dT}{dt} = c^2 T \frac{d^2 X}{dx^2}$$

$$\Rightarrow \frac{1}{c^2 T} \frac{dT}{dt} = \frac{1}{X} \frac{d^2 X}{dx^2} = -p^2$$

Solving,

$$\frac{1}{c^2 T} \frac{dT}{dt} = -p^2$$

$$\Rightarrow \frac{dT}{dt} + p^2 c^2 T = 0$$

$$\Rightarrow (D + p^2 c^2) T = 0$$

$$\Rightarrow D + p^2 c^2 = 0$$

$$\Rightarrow D = -p^2 c^2$$

$$T = C_1 e^{-p^2 c^2 t}$$

Again,

$$\frac{1}{X} \cdot \frac{dX^2}{dx^2} = -p^2$$

$$\Rightarrow \frac{dX^2}{dx^2} + p^2 X = 0$$

$$\Rightarrow (D^2 + p^2) X = 0$$

$$\Rightarrow D^2 + p^2 = 0$$

$$\Rightarrow D = \pm ip$$

$$X = C_2 \cos px + C_3 \sin px$$

Putting the values of X & T in eqⁿ. (2),

$$u = C_1 e^{-p^2 z^2 t} (C_2 \cos px + C_3 \sin px).$$

Q.1) Find the temperature in a bar of length 2 whose ends kept at zero and lateral surface insulated, if the initial temperature is $\sin\left(\frac{\pi x}{2}\right) + 3 \sin\left(\frac{5\pi x}{2}\right)$.

Solⁿ:

The one dimensional heat flow eqⁿ. is,

$$\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2} \quad \text{--- (1)}$$

The solution of eqⁿ. (1) is,

$$u = (C_1 \cos px + C_2 \sin px) C_3 e^{-c^2 p^2 t} \quad \text{--- (2)}$$

The boundary conditions are:-

$$u = 0, x = 0$$

$$u = 0, x = 2$$

$$u = \sin\left(\frac{\pi x}{2}\right) + 3 \sin\left(\frac{5\pi x}{2}\right), t = 0$$

Applying condition $u = 0$ & $x = 0$ in eqⁿ. (2),

$$C_1 = 0$$

Putting this value of C_1 in eqⁿ. (2),

$$u = C_2 C_3 \sin px \cdot e^{-c^2 p^2 t} \quad \text{--- (3)}$$

Applying condition $u=0$ & $x=2$ in eqⁿ. (3),

$$\begin{aligned}\sin 2p &= 0 \\ \Rightarrow \sin 2p &= \sin n\pi \\ \Rightarrow 2p &= n\pi \\ \Rightarrow p &= \frac{n\pi}{2}\end{aligned}$$

Putting this value of p in eqⁿ. (3),

$$u = C_2 C_3 \sin\left(\frac{n\pi x}{2}\right) \cdot e^{-\left(\frac{n^2 \pi^2 c^2 t}{4}\right)} \quad \text{--- (4)}$$

$$\Rightarrow u = b_n \sin\left(\frac{n\pi x}{2}\right) \cdot e^{-\left(\frac{n^2 \pi^2 c^2 t}{4}\right)}$$

The most general solution is,

$$u = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{2}\right) \cdot e^{-\left(\frac{n^2 \pi^2 c^2 t}{4}\right)} \quad \text{--- (5)}$$

Applying condition $u = \sin\left(\frac{\pi x}{2}\right) + 3 \sin\left(\frac{5\pi x}{2}\right)$ & $t=0$,
in eqⁿ. (5),

$$\sin\left(\frac{\pi x}{2}\right) + 3 \sin\left(\frac{5\pi x}{2}\right) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{2}\right)$$

$$\begin{aligned}\Rightarrow \sin\left(\frac{\pi x}{2}\right) + 3 \sin\left(\frac{5\pi x}{2}\right) &= b_1 \sin\left(\frac{\pi x}{2}\right) + b_2 \sin\left(\frac{2\pi x}{2}\right) \\ &+ \dots + b_5 \sin\left(\frac{5\pi x}{2}\right) + \dots\end{aligned}$$

Equating the coefficients of both sides,

$$b_1 = 1, b_2 = b_3 = b_4 = 0, b_5 = 3, b_6 = \dots = 0$$

Putting these values in eqⁿ. (5),

$$u = \sin\left(\frac{\pi x}{2}\right) e^{-\frac{\pi^2 c^2 t}{4}} + 3 \sin\left(\frac{5\pi x}{2}\right) e^{-\frac{25\pi^2 c^2 t}{4}}.$$

Q.2) Solve the equation, $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$ with boundary

condition $u(x, 0) = 3 \sin n\pi x$, $u(0, t) = 0$,
 $u(l, t) = 0$, where $0 < x \leq l$.

Solⁿ:

Given, $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$ ——— (1)

The solution of eqⁿ. (1) is,

$$u = C_1 e^{-p^2 t} (C_2 \cos px + C_3 \sin px) \text{ ——— (2)}$$

Applying the condition $u = 0$ & $x = 0$ in eqⁿ. (2),

$$C_2 = 0$$

Putting this value of C_2 in eqⁿ. (2),

$$u = C_1 C_3 e^{-p^2 t} \sin px \text{ ——— (3)}$$

Applying the condition $u=0$ & $x=l$ in eqⁿ. (3),

$$\begin{aligned}\sin pl &= 0 = \sin n\pi \\ \Rightarrow pl &= n\pi \\ \Rightarrow p &= \frac{n\pi}{l}\end{aligned}$$

Putting this value of p in eqⁿ. (3),

$$u = C_1 C_3 \cdot e^{-\left(\frac{n^2 \pi^2 t}{l^2}\right)} \cdot \sin\left(\frac{n\pi x}{l}\right) \text{ ————— (4)}$$

The most general solution is,

$$u = \sum_{n=1}^{\infty} b_n e^{-\left(\frac{n^2 \pi^2 t}{l^2}\right)} \cdot \sin\left(\frac{n\pi x}{l}\right) \text{ ————— (5)}$$

Applying the condition $u=3 \sin(n\pi x)$ & $t=0$ in eqⁿ. (5),

$$3 \sin(n\pi x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right)$$

$$\Rightarrow B_n = 3 \text{ \& } l=1$$

Putting in eqⁿ. (5), we get,

$$u(x, t) = 3 \sin(n\pi x) e^{-n^2 \pi^2 t} \text{ ————— (6)}$$

Two Dimensional Heat Flow

Consider the heat flow in a metal plate of uniform thickness. The flow occurs in the direction parallel to the length 'x' and breadth 'y' of the plate. We assume there is no heat flow along the normal to the rectangular element ABCD.

Let $u(x, y, t)$ be the temperature at any point (x, y) of the plate at time t , then the equation is,

$$\frac{\partial u}{\partial t} = c^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) \quad \text{--- (1)}$$

In a steady state, the temperature u is independent of time 't'. Therefore, $\frac{\partial u}{\partial t} = 0$

and the eqⁿ. (1) becomes,

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \quad \text{--- (2)}$$

This is called Laplace Equation in two dimensions.

Q.1) Solve: $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$, which satisfies the conditions:- $u(0, y) = u(l, y) = u(x, 0) = 0$ and $u(x, a) = \sin\left(\frac{n\pi x}{l}\right)$.

Solⁿ:

The given eqⁿ. is,

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \quad \text{--- (1)}$$

Let $u = XY$ --- (2) be a solution of eqⁿ. (1)

$$\frac{\partial^2 u}{\partial x^2} = Y \frac{d^2 X}{dx^2}$$

$$\frac{\partial^2 u}{\partial y^2} = X \frac{d^2 Y}{dy^2}$$

Putting these in eqⁿ. (1),

$$Y \frac{d^2 X}{dx^2} + X \frac{d^2 Y}{dy^2} = 0$$

$$\Rightarrow \frac{1}{X} \frac{d^2 X}{dx^2} = -\frac{1}{Y} \frac{d^2 Y}{dy^2} = -p^2$$

Solving,

$$\frac{1}{X} \frac{d^2 X}{dx^2} = -p^2$$

$$\Rightarrow (D^2 X + p^2 X) = 0$$

$$\Rightarrow D^2 + p^2 = 0$$

$$\Rightarrow D = \pm ip$$

$$X = C_1 \cos px + C_2 \sin px$$

Also solving,

$$-\frac{1}{Y} \frac{d^2 Y}{dy^2} = -p^2$$

$$\Rightarrow D^2 Y - p^2 Y = 0$$

$$\Rightarrow D^2 - p^2 = 0$$

$$\Rightarrow D = \pm p$$

$$Y = C_3 e^{py} + C_4 e^{-py}$$

Putting the value of X & Y in eqⁿ. (2),

$$u = (C_1 \cos px + C_2 \sin px) (C_3 e^{py} + C_4 e^{-py}) \text{---(3)}$$

Applying condition $u=0$ & $x=0$ in eqⁿ. (3),

$$C_1 = 0$$

Putting value of C_1 in eqⁿ. (3),

$$u = C_2 \sin px (C_3 e^{py} + C_4 e^{-py}) \text{---(4)}$$

Applying condition $u=0$ & $x=l$ in eqⁿ. (4),

$$\begin{aligned}\sin p l &= 0 \\ \Rightarrow \sin p l &= \sin n \pi = \\ \Rightarrow p l &= n \pi \\ \Rightarrow p &= \frac{n \pi}{l}\end{aligned}$$

Putting this value of p in eqⁿ. (4),

$$u = C_2 \sin\left(\frac{n \pi x}{l}\right) \cdot \left(C_3 e^{\frac{n \pi y}{l}} + C_4 e^{-\frac{n \pi y}{l}} \right) \quad (5)$$

Applying condition $u=0$ & $y=0$ in eqⁿ. (5),

$$\begin{aligned}C_3 + C_4 &= 0 \\ \Rightarrow C_4 &= -C_3\end{aligned}$$

Putting this value of C_4 in eqⁿ. (5),

$$u = C_2 C_3 \cdot \sin\left(\frac{n \pi x}{l}\right) \left(e^{n \pi y / l} - e^{-\frac{n \pi y}{l}} \right) \quad (6)$$

Applying condition $u = \sin\left(\frac{n \pi x}{l}\right)$ & $y=a$ in eqⁿ. (6),

$$\sin\left(\frac{n \pi x}{l}\right) = C_2 C_3 \cdot \sin\left(\frac{n \pi x}{l}\right) \cdot \left(e^{\frac{n \pi a}{l}} - e^{-\frac{n \pi a}{l}} \right)$$

Equating the coefficients of $\sin\left(\frac{n\pi x}{l}\right)$,

$$C_2 C_3 \left(e^{\frac{n\pi a}{l}} - e^{-\frac{n\pi a}{l}} \right) = 1$$

$$\Rightarrow C_2 C_3 = \frac{1}{\left(e^{\frac{n\pi a}{l}} - e^{-\frac{n\pi a}{l}} \right)}$$

Putting this value of $C_2 C_3$ in eqⁿ. (6),

$$u = \sin\left(\frac{n\pi x}{l}\right) \left(\frac{e^{\frac{n\pi y}{l}} - e^{-\frac{n\pi y}{l}}}{e^{\frac{n\pi a}{l}} - e^{-\frac{n\pi a}{l}}} \right)$$

$$\Rightarrow u = \sin\left(\frac{n\pi x}{l}\right) \left\{ \frac{\sinh\left(\frac{n\pi y}{l}\right)}{\sinh\left(\frac{n\pi a}{l}\right)} \right\} \text{ ————— (7)}$$

$$\left[\sinh x = \frac{e^x - e^{-x}}{2} \right]$$

Q.2) A rectangular plate with insulated surfaces is 10 cm wide and so long compared to its width that it may be considered infinite in length without introducing an appreciable error. If the temperature along the short edge $y=0$ is given by:-

$$u(x, 0) = 20x, \quad 0 < x \leq 5$$

$$u(x, 0) = 20(10-x), \quad 5 < x < 10$$

while the two long edges $x=0$ and $x=10$ as well as the other short edges are kept at 0°C . Find the steady state temperature at any point (x, y) of the plate?

Solⁿ:

In the steady state,

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \quad \text{--- (1)}$$

The given boundary conditions:-

$$u = 0 \quad \& \quad x = 0$$

$$u = 0 \quad \& \quad x = 10$$

$$u = 0 \quad \& \quad y \rightarrow \infty$$

$$u = 20x, \quad 0 < x \leq 5 \quad \& \quad y = 0$$

$$u = 20(10-x), \quad 5 < x < 10 \quad \& \quad y = 0$$

The solution of eqⁿ. (1) is,

$$u = (C_1 \cos px + C_2 \sin px) (C_3 e^{py} + C_4 e^{-py}) \quad \text{--- (2)}$$

Applying condition $u=0$ & $x=0$ in eqn. (2),

$$C_1 = 0$$

Putting this value of C_1 in eqn. (2),

$$u = C_2 \sin px (C_3 e^{py} + C_4 e^{-py}) \quad \text{--- (3)}$$

Applying condition $u=0$ & $x=10$ in eqn. (3),

$$\sin 10p = 0$$

$$\Rightarrow \sin 10p = \sin n\pi$$

$$\Rightarrow 10p = n\pi$$

$$\Rightarrow p = \frac{n\pi}{10}$$

Putting this value of p in eqn. (3),

$$u = C_2 \sin\left(\frac{n\pi x}{10}\right) (C_3 e^{\frac{n\pi y}{10}} + C_4 e^{-\frac{n\pi y}{10}}) \quad \text{--- (4)}$$

Applying condition $u=0$ & $y \rightarrow \infty$ in eqn. (4),

$$0 = C_2 \sin\left(\frac{n\pi x}{10}\right) (C_3 e^{\infty} + 0)$$

$$\Rightarrow C_3 = 0$$

Putting this value of C_3 in eqn. (4),

$$u = C_2 C_4 \sin\left(\frac{n\pi x}{10}\right) e^{-\frac{n\pi y}{10}} \quad \text{--- (5)}$$

$$\Rightarrow u = b_n \sin\left(\frac{n\pi x}{10}\right) \cdot e^{\left(\frac{-n\pi y}{10}\right)} \quad \text{--- (6)}$$

The most general solution is,

$$u(x, y) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{10}\right) e^{\left(\frac{-n\pi y}{10}\right)} \quad \text{--- (7)}$$

Applying condition $u = 20x$, $0 < x \leq 5$
 $u = 20(10-x)$, $5 < x < 10$
 & $y = 0$ in eqⁿ. (7),

$$b_n = \frac{2}{10} \int_0^{10} u(x, 0) \sin\left(\frac{n\pi x}{10}\right) dx$$

[Using Fourier Sine Series]

$$b_n = \frac{2}{10} \int_0^5 \underbrace{20x}_{\text{I}} \underbrace{\sin\left(\frac{n\pi x}{10}\right)}_{\text{II}} dx + \frac{2}{10} \int_5^{10} \underbrace{20(10-x)}_{\text{I}} \underbrace{\sin\left(\frac{n\pi x}{10}\right)}_{\text{II}} dx$$

$$\Rightarrow b_n = 4 \left\{ \left[x \left(\frac{-\cos \frac{n\pi x}{10}}{n\pi/10} \right) \right]_0^5 - \int_0^5 (1) \left(\frac{-\cos \frac{n\pi x}{10}}{n\pi/10} \right) dx \right\} \\ + 4 \left\{ \left[(10-x) \left(\frac{-\cos \frac{n\pi x}{10}}{n\pi/10} \right) \right]_{5}^{10} - \int_5^{10} (-1) \left(\frac{-\cos \frac{n\pi x}{10}}{n\pi/10} \right) dx \right\}$$

[Using Integration By Parts]

$$\begin{aligned}
 \Rightarrow b_n &= 4 \left[-x \cdot \frac{\cos\left(\frac{n\pi x}{10}\right)}{\frac{n\pi}{10}} + \frac{\sin\left(\frac{n\pi x}{10}\right)}{\left(\frac{n\pi}{10}\right)^2} \right]_0^5 \\
 &\quad + 4 \left[(10-x) \frac{\cos\left(\frac{n\pi x}{10}\right)}{\frac{n\pi}{10}} - \frac{\sin\left(\frac{n\pi x}{10}\right)}{\left(\frac{n\pi}{10}\right)^2} \right]_5^{10} \\
 &= 4 \left[\frac{-5 \cos\left(\frac{n\pi}{2}\right)}{\frac{n\pi}{10}} + \frac{\sin\left(\frac{n\pi}{2}\right)}{\left(\frac{n\pi}{10}\right)^2} + 0 - 0 \right] \\
 &\quad + 4 \left[0 - 0 + 5 \frac{\cos\left(\frac{n\pi}{2}\right)}{\frac{n\pi}{10}} + \frac{\sin\left(\frac{n\pi}{2}\right)}{\left(\frac{n\pi}{10}\right)^2} \right] \\
 &= \frac{4}{\left(\frac{n\pi}{10}\right)^2} \left(2 \sin \frac{n\pi}{2} \right)
 \end{aligned}$$

$$b_n = \frac{800}{n^2 \pi^2} \sin\left(\frac{n\pi}{2}\right)$$

If n is even (i.e. $n=2, 4, 6, \dots$), then $\sin\left(\frac{n\pi}{2}\right) = 0$
 $\Rightarrow b_n = 0$

If n is odd (i.e. $n=1, 3, 5, \dots$), then,

$$b_n = \pm \frac{800}{n^2 \pi^2}$$

$$\text{i.e. } b_n = \frac{(-1)^{n+1} 800}{(2n-1)^2 \pi^2} \quad [n \text{ is odd}]$$

Eqⁿ. (7) becomes,

$$u(x, y) = \frac{800}{\pi^2} \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{(2n-1)^2} \frac{\sin \frac{(2n-1)\pi x}{10}}{10} \cdot e^{-\frac{(2n-1)\pi y}{10}}.$$